

Jae Yun

US Citizen, No Sponsorship Required

San Diego, California

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Education

UC San Diego

09/2024 - Present

M.S. Computer Science (AI/ML)

GPA: 4.0/4.0

- Relevant courses: Fairness and Bias in ML, LLM Topics, Web Mining, Internet Safety with LLMs.

UC Santa Barbara

09/2019 - 09/2024

B.S. Statistics & Data Science, B.A. Geographical Information Science (GIS)

GPA: 3.73/4.0

- Relevant courses: Big Data Analytics, ML Applications, ML Theory, Statistics/Probability, A/B Testing, Time Series.

Work Experience

Data Scientist

10/2024 - 02/2024

Volpe National Transportation Systems Center

- Optimized SQL database performance by 25% for 20M+ observations through parallel processing and memory optimization techniques in R.
- Developed an automated continuous integration testing platform with 50+ test cases using GitHub Actions, reducing bug reports by 80% and 10+ hours of testing time per release.
- Built an automated binary-search optimization tool in Python for supply chain routing, reducing processing time by 90%.
- Maintain interactive dashboards with Tableau and ArcGIS Pro and establish regular communication with external-facing sponsors.
- Collaborating on GitHub with a team of 9 developers, managing version control through Git.

Data Science Intern

06/2024 - 10/2024

Volpe National Transportation Systems Center

- Functionalized and optimized R scripts for our ship emissions calculation model, integrating SQL database queries for CO₂ emissions calculations.
- Built Python scripts to parse log files (TXT, XML, CSV) to visualize convergence times and system outputs.
- Modularized Jupyter Notebook workflows into reusable Python scripts, reducing manual adjustments by 50%.
- Built PowerBI dashboards from Excel templates for sponsor decision-making, integrating multiple data sources.

Statistics Undergraduate Teaching Assistant

09/2022 - 03/2024

UCSB Department of Statistics

- Developed and deployed a class scheduling tool using HTML/CSS and JavaScript, reducing administrative workload by 20+ hours per quarter. Available at: github.com/jaewoongy/class-scheduler.
- Lead sections and office hours and assisted in course development for ML and statistics courses.

Relevant Projects

Undergraduate Research Assistant - Machine Learning

06/2022 - 06/2023

Dr. Linda Petzold, UCSB Computer Science Department

- Co-authored and published research at the PLOS conference. Published work available at: [arXiv:2308.10892](https://arxiv.org/abs/2308.10892).
- Used PyTorch for Bayesian neural networks to assess the uncertainty of current algorithm point estimates
- Validated models using machine learning algorithms (Laplace, Bayesian Linear Regression) to ensure neural network reliability.

Academic Research Study - Time Series Modeling

01/2023 - 06/2023

- Utilized SARIMA time-series models and techniques in R for data pre-processing and cleaning, parameter selection, model forecasting, and evaluation
- Research findings and methodology are documented in more detail available [here](#).

Skills

- Python (Pytorch, Pyspark, Scikit-Learn), SQL, Google Cloud Platform, R, Tableau, Power BI, ArcGIS Pro, HTML/CSS, JavaScript, Git Version Control